BE (E&TC)	PLC SCADA and Automation LAB	
Experiment No: 5	**Counting Objects**	**Pages 7-9**

AIM: To study of Counting Objects (Any Application of Counter using PLC)

Examples:1. Bottle Filling & Packaging – Count bottles and stop after a preset number for each carton.

2. Parking system

Objective:

- To understand the working of counters in a PLC.
- To count objects passing through a sensor using a PLC.
- To control output devices based on the counted objects.

Theory:

A **PLC Counter** instruction counts the number of events (e.g., passing objects) detected by an input device such as a sensor.
Counters can be:

- **Up Counter (CTU)** – increments the count each time the input is activated.
- **Down Counter (CTD)** – decrements the count.
- **Up/Down Counter (CTUD)** – can count in both directions.

In this experiment, an **Up Counter** is used to keep track of the number of objects passing on a conveyor belt. A proximity sensor generates a pulse each time an object passes, which is fed to the PLC input. The PLC increments the counter value and displays it on the HMI or activates an output after a set number of objects.

Draw Ladder Diagram for Bottle filling plant:

CONCLUSION

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QUESTIONS:

1. What is a counter in PLC programming?
2. Differentiate between **up counter** and **down counter** in PLC.
3. What is the function of the **preset value (PV)** in a PLC counter?
4. Explain how a counter is reset in PLC.
5. Why are counters important in industrial automation?